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Hybrid synaptic structure for spiking neural network realization

S Razmkhah[1,2,*], **M A Karamuftuoglu**[1,2] and **A Bozbey**[2]

[1] Ming Hsieh Department of Electrical and Computer Engineering, University of Southern California, Los Angeles, CA 90089 United States of America
[2] Department of Electrical and Electronics Engineering, TOBB University of Economics and Technology, Ankara, Turkey

E-mail: razmkhah@usc.edu

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Abstract
Neural networks and neuromorphic computing represent fundamental paradigms as alternative approaches to Von-Neumann-based implementations, advancing in the applications of deep learning and machine vision. Nonetheless, conventional semiconductor circuits encounter challenges in achieving ultra-fast processing speed and low power consumption due to their dissipative properties. Conversely, single flux quantum circuits exhibit inherent spiking behavior, showcasing their characteristics as a promising candidate for spiking neural networks (SNNs). In this work, we present a compact hybrid synapse circuit to mimic the biological interconnect functionality, enabling the weighting operations for excitatory and inhibitory impulses. Additionally, the proposed structure facilitates input accumulation, which is performed before the activation function. In the experiments, our synaptic structure interfaces with a soma circuit fabricated using a commercial Nb process, underscoring its compatibility and supporting its potential for integration into efficient neural network architectures. The weight value on the synapse is configurable by utilizing cryo-CMOS circuits, providing adaptability to the inference networks. We've successfully designed, fabricated, and partially tested the JJ-Synapse within our cryocooler system, enabling high-speed inference implementation for SNNs.

Keywords: artificial synapse, spiking neural network, single flux quantum, superconductor electronics

1. Introduction

Modern data management and processing demands the computational power to handle vast volumes of complex and interconnected information. Classical algorithms often prove inadequate for such tasks, necessitating the utilization of machine learning and deep learning algorithms for training computers to perform these operations effectively [1–3]. While users typically employ conventional CPU and GPU architectures for deep learning and network training, the increasing reliance on this technology underscores the need for faster and more power-efficient computational resources [4, 5].

Neural networks and neuromorphic architectures play pivotal roles in efficiently implementing learning algorithms in machines. By emulating the functionality of the biological brain, neuromorphic computers offer an alternative computational approach and a robust avenue for simulating artificial intelligence [6, 7]. Several neuromorphic architecture implementations, such as Intel's Loihi and Loihi2 chips designed for machine learning acceleration [3], rely on conventional CMOS logic. These designs integrate memory into

* Author to whom any correspondence should be addressed.

the arithmetic logic unit, enabling in-memory computing and significantly enhancing the speed and robustness of the learning process. However, they continue to use voltage-level logic to mimic neural processes.

In contrast, research groups and companies, including IBM and Intel, explore innovative devices like memristors to replicate the learning and path formation observed in biological brain synapses [8, 9]. Memristors change their impedance as current flows through them, exhibiting behavior analogous to changes in ion channels and resistance drops akin to synapses [10, 11]. Nevertheless, memristors are associated with substantial power consumption, and their recovery time falls within the millisecond range, rendering them comparatively slow.

Superconductor electronics have long been regarded as a promising candidate for neural networks, with researchers exploring their potential for many years [12–21]. In [13], the authors introduce spiking neurons with standard superconductor components and a SQUID structure. Furthermore, the designs in [16, 17, 19] extend the research on bio-inspired neurons and synapses with superconductor nanowires. The study in [20] demonstrates the utilization of multi-fluxon storage cells [22] for arbitrary pulse generation on synapses. As discussed in [14], the synaptic connections are realized by magnetic Josephson junctions (MJJs).

Single flux quantum (SFQ) logic, a key component of superconductor electronics, can operate at frequencies in the tens of gigahertz range while consuming power levels three orders of magnitude lower than traditional CMOS [23–25]. Moreover, the pulse-based logic inherent in SFQ is akin to biological neural signals, where the presence of a pulse represents logic one, and its absence denotes logic zero. SFQ cells are also inherently synchronous, relying on clock signals, with memory integrated into the logic. Although creating dense memory structures in SFQ can be challenging [26–29], it is exceptionally well-suited for in-memory computation. The combination of neuromorphic architecture and SFQ technology is advantageous.

This work introduces a synapse structure called JJ-Synapse, leveraging a hybrid superconductor-semiconductor technology. Due to the quantized nature of SFQ pulses, the weights in the artificial JJ-Synapse are also quantized, necessitating training in a quantized manner. While this may slightly reduce algorithmic accuracy, it offers substantial advantages regarding on-chip memory and processing speed [30, 31].

The emerging superconductor devices such as a three-terminal superconducting nanowire device nTron [32] and MJJs [14] may encounter limitations on the reliability, which is ongoing research. Although SFQ technology offers remarkable capabilities, it is not self-sufficient for implementing a fully functional, high-performance neuromorphic processor. Resources such as biasing for individual neurons and storage for training parameters during execution are still required. To address this, we propose an SFQ-CMOS hybrid synapse cell that capitalizes on the strengths of each technology. CMOS components provide the necessary biases for JJ-Synapses and store weight values for the network in MOSFET switches. SiGe FETs are compatible with 4 K operation, as shown by various groups [33]. In this work, we designed cryo-CMOS SG13G2 using the IHP 0.13 μm BiCMOS process. Since the CMOS only switches when configuring weights and remains fixed during inference operations, it does not impose significant speed limitations and has a negligible impact on power consumption.

The proposed synapse design incorporates negative and positive weights. It is interfaced with a quantizer/buffer circuit to propagate the accumulated post-synaptic result into the JJ-Soma cell [15], which performs a somatic activation function and is compatible with the proposed structure. The whole integration, comprising JJ-Synapse, JJ-Soma, and a buffer, will mimic a neuron with an integrate-and-fire neuron model and with its synaptic connections. The resulting JJ-Neuron is expected to operate at nearly 20 GHz and maintain compatibility with SFQ logic, including RSFQ. This JJ-Neuron structure is a pivotal component for our future work, enabling the realization of a complete and fully functional neural network.

2. Model and methodology

A crucial aspect of implementing an artificial neural network is the presence of synapses with programmable weights that can accommodate diverse input levels, all while maintaining reasonable circuit size, speed, and power consumption. Moreover, these synapses should be versatile enough to process excitatory and inhibitory inputs. Given that our fundamental data unit is the SFQ pulse, the synapse must have the capability to generate a specific number of pulses, as determined by equation (1), at its output,

$$u = \sum_{k=1}^{n} \left(w_k P_k - w_k' N_k \right). \tag{1}$$

Since the operating voltage is on the positive level, we utilize the direction of input to achieve the summation of positive and negative inputs. For this reason, we treat the positive and negative branches separately and reflect equivalent hardware operations into the corresponding equation. In equation (1), the variables are defined as follows: k represents the synaptic input port index, n is the total number of synapses, w and w' are the positive and negative weights, N and P correspond to the presence of presynaptic input taking the values of 0 or 1, and u signifies the accumulated output of the JJ-Synapse. The overall operation corresponds to the MAC (Multiply-Accumulate) module, which has higher hardware costs on superconductor circuit design methodology for neural network applications.

In our design, SFQ pulses from presynaptic connections are accumulated in both positive and negative branches, and then the two values are summed up. The value u is provided to the JJ-Soma [15], which acts as an activation function. The main component of such a comparator is a SQUID with an interrupted resistor. The applied input will be temporarily stored within this SQUID-like structure if the SQUID's junction does not switch. When the input u exceeds the predefined threshold, an SFQ pulse is generated at the output of JJ-Soma. Figure 1 illustrates the overall implementation of the model utilizing the synapse structure.

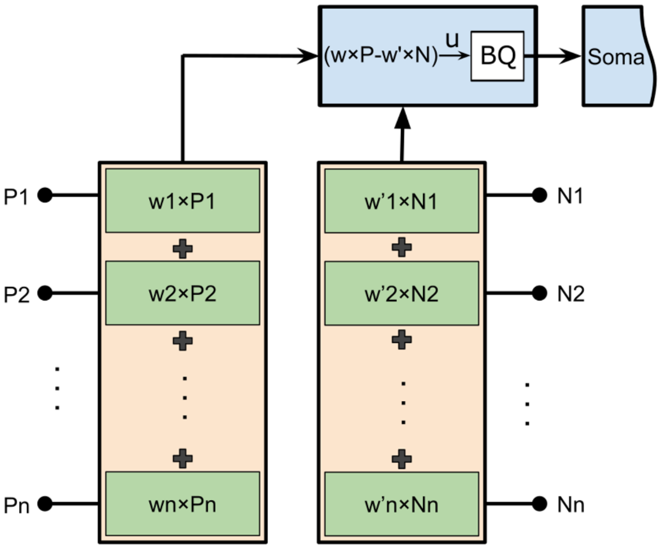

Figure 1. Block diagram of the JJ-Synapse with positive and negative inputs. The weighted inputs are summed and propagated to JJ-Soma by using the Buffer/Quantizer (BQ) circuit.

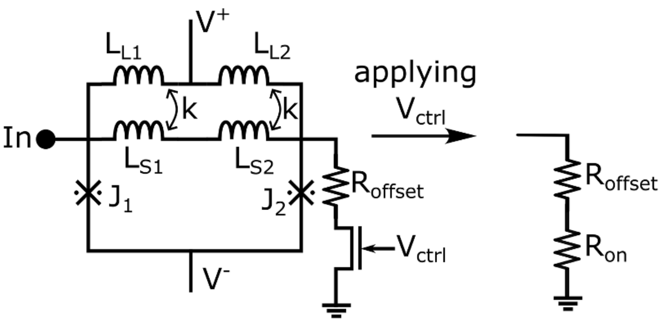

Figure 2. The synapses will use the drain-source resistance of the MOSFET for switching. If the $R_{\mathrm{on}} + R_{\mathrm{offset}} \simeq 2.2\ \Omega$ then the SQUID will switch and generate SFQ pulse.

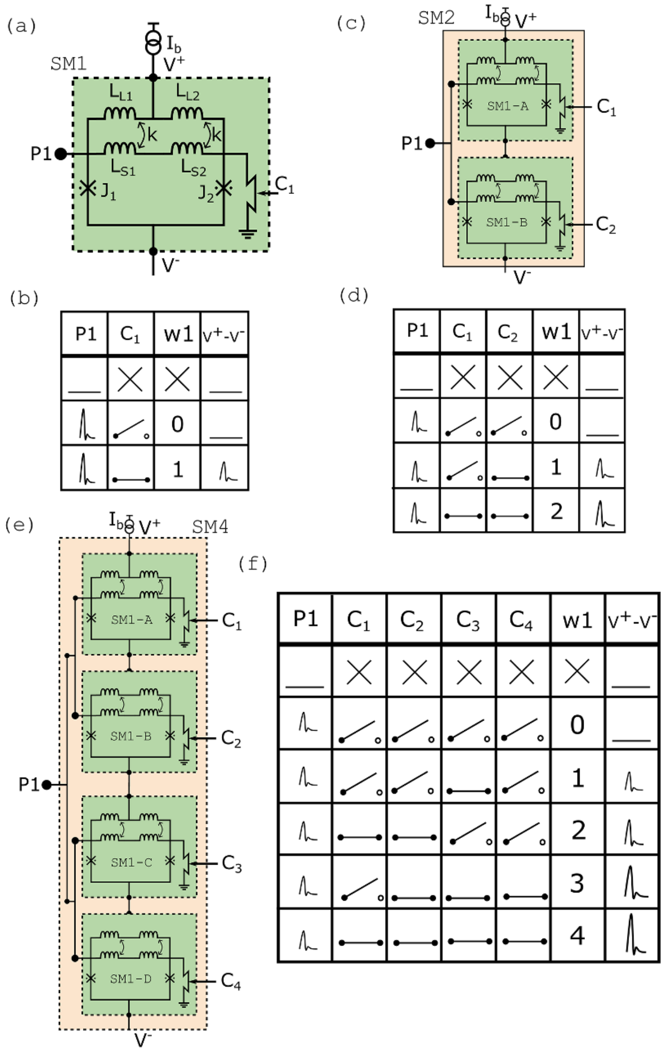

Figure 3. Design of the building block for the synapse circuit. To overlap multiple input ports or increase the synaptic weight without losing any operating speed, we use a series of SQUID. (a) Demonstration of a single cell. (b) The truth table for a single cell as it switches. (c) Two connected cells make a weight of two for a single input. (d) The truth table for two cells. (e) Two connected cells of size two for a single input that can achieve weight four. (f) The truth table for *SM*4 unit. The symbol *X* in the tables denotes the *do not care* condition, and any configuration set of *C* can be used interchangeably to achieve different weights. In the case of all switches being off, the circuit will be open, and there will be no current flow. As a result, a zero weight value will be achieved.

To align this model with SFQ and facilitate hardware integration, we assume that P_k and N_k are equal to one with the presence of an SFQ pulse and zero for the absence of a pulse. We map the range of w_k and w_k' to integers between 0 and 4 to demonstrate a sample structure that concatenates multiple cells to achieve bigger synaptic weights. As previously mentioned, imposing these limits may reduce network accuracy but can significantly enhance operating speed and simplify hardware integration. Our synapse circuit comprises serially connected SQUID loops and input *LR* circuits, each coupled with a SQUID loop and matched with the outputs of the SFQ library elements, as depicted in figure 3 for the positive branch. The building blocks (SM1) of the synapse circuit are serially connected, significantly reducing bias current and power consumption. Each synapse circuit accepts spiking inputs and generates the positive or negative weighted output for the following layer. The switches $C_1 - C_4$ shown in figure 3 are intended to be implemented using cryo-CMOS transistors. The transistor connection to the SQUIDs is shown in figure 2. Figure 3(a) illustrates the circuit schematics of SM1. An applied SFQ pulse at the P1 input propagates through L_{S1} and L_{S2}, which are coupled to L_{L1} and L_{L2}, respectively. This incoming pulse is coupled through these inductors, triggering the SQUID loop. Josephson junctions J_1 or J_2 switch and generate an SFQ pulse between nodes $V+$ and $V-$. Consequently, the circuit achieves the case where the weight $w_1 = 1$ with the input $P_1 = 1$. When the C_1 switch is open, current cannot flow from L_{S1} and L_{S2}, resulting in a zero voltage difference on the result nodes. Thus, the circuit achieves a zero weight ($w_1 = 0$) in this configuration. The first row in figure 3(b) corresponds to the absence of the input pulse, and there will be no voltage difference observed. For such cases, the weight value does not affect the result.

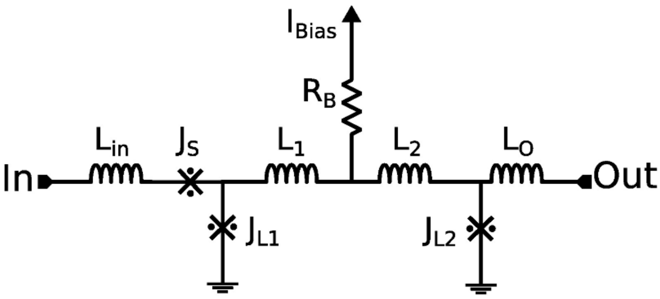

Figure 4. The Buffer/Quantizer (BQ) circuit functions similarly to the digital SQUID. BQ will sum the negative and positive fluxes applied to its loop and then provide quantized SFQ pulses based on the change in the flux. ($Lin = 0.8$ pH, $L0 = 1.323$ pH, $L1 = L2 = 3.91$ pH, $JS = 133$ μA, $JL1 = 112$ μA, $JL2 = 189$ μA, $RB = 8.5$ Ω).

We can generate weighted inputs by serially connecting these SQUID loops and providing the SFQ pulses simultaneously to each loop of SM #X stages. For instance, by connecting two in series, we can implement weights 1 and 2 by controlling C_1 and C_2 based on the table shown in figure 3(d) for SM2. When $C_1 = 0$ and $C_2 = 1$, the voltage difference at the output nodes equals one SFQ amplitude, resulting in a weight of one ($w_1 = 1$). Similarly, when $C_1 = 1$ and $C_2 = 1$, the difference equals two SFQ amplitudes ($w_1 = 2$). In addition to the weighted inputs, if the network requires multiple input ports (P_k), multiple building blocks of SM1 circuits can be serially connected to establish a structure with a high-fanin feature. Based on the table shown in figure 3(f), it is possible to have w_1 with zero to four integer weights on a single input port. A comparator circuit inspires the buffer/Quantizer (BQ) circuit, which acts similarly to an asynchronous quasi-one junction SQUID circuit. Figure 4 demonstrates the schematic of the BQ circuit. This circuit converts the cumulative energy of multiple pulses generated by SM #X stages into several fast pulses proportional to $\sum P$, suitable for JJ-Soma input. Escape junctions J_S are used to eliminate the SFQ backfiring. We then connect the BQ circuit to the JJ-Soma for activation function. For a more accurate implementation of the neural network, it is essential that the synapses can handle negative weights. In figure 5, we have designed a circuit connecting two SMX circuits in a current-differential configuration. In this setup, the sum of positive weighted inputs ($\sum P$) and the sum of negative weighted inputs ($\sum N$) are differentiated to determine the net weighted inputs ($\sum P - \sum N$). Consequently, the current passing through the L_{NT} inductor is directly proportional to the difference between $\sum P$ and $\sum N$ currents. The analog current value, K times this difference ($K \times (\sum P - \sum N)$, where K is the coupling factor), is converted into a quantized spiking configuration using the BQ circuit.

On the negative side of the circuit, a matching impedance, Z_{M}, is employed to match the impedance seen by the L_{NT} and L_{PT} inductors, ensuring that the positive and negative sides remain symmetrical and balanced. This configuration allows the synapse to handle both positive and negative weights effectively.

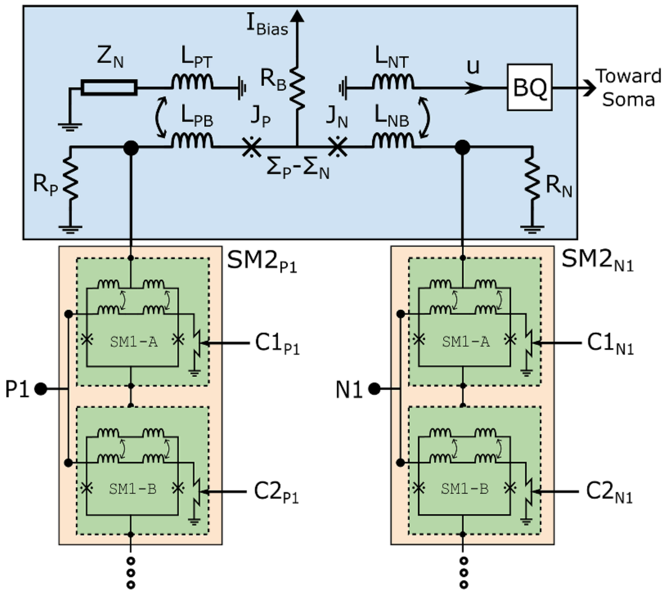

Figure 5. Schematic of the JJ-Synapse with positive and negative inputs. An equivalent load of the Buffer/Quantizer (BQ) circuit is attached to the left-hand side for symmetry. (SM1: $L_{L1} = L_{L2} = L_{S1} = L_{S2} = 30.12$ pH $J0 = J1 = 109$ μA. Accumulator: $L_{\mathrm{PT}} = L_{\mathrm{NT}} = 8$ pH, $L_{\mathrm{PB}} = L_{\mathrm{NB}} = 5.8$ pH, $JP = JN = 101$ μA, $RP = RN = 0.98$ Ω).

3. Simulation results

We conducted simulations of the circuits using the JSIM program, followed by an optimization process to enhance parameter margins. This optimization was carried out using a tool that employs the particle swarm optimization algorithm developed by our team [34]. The main objective of this optimization was to maximize the parameter margins of the circuit, making it robust against variations in the fabrication process. As a result of this optimization, we achieved parameter margins exceeding 20%, which ensures the circuit's reliability even with manufacturing process variations. The bias margin, while relatively small at less than 5%, is not a concern as it can be externally applied and finely adjusted for the circuit.

We also simulated the JJ-Synapse structure depicted in figure 3(e). Eight SQUID cells are required to achieve 3-bit weight values. These eight SQUIDs can be selectively turned on or off to achieve the desired weight value. Figure 6 presents the input signal and output of the JJ-Synapse for different weight values. It's worth noting that increasing the weight value results in higher signal amplitudes, increasing the energy. The applied input SFQ pulse had a frequency of 10 GHz, but it can be further increased to ≈25 GHz for a synapse with 3-bit adjustable weight capability.

We performed simulations of the circuit with various JJ-Soma thresholds and different weight values. The overall operation is synchronized by using DFFs. The results of these simulations are presented in figure 7, which showcases the changes in circuit phases. Since each fluxon induces a 2π phase change, the values are divided by 2π for clarity. This figure measured negative and positive values from the

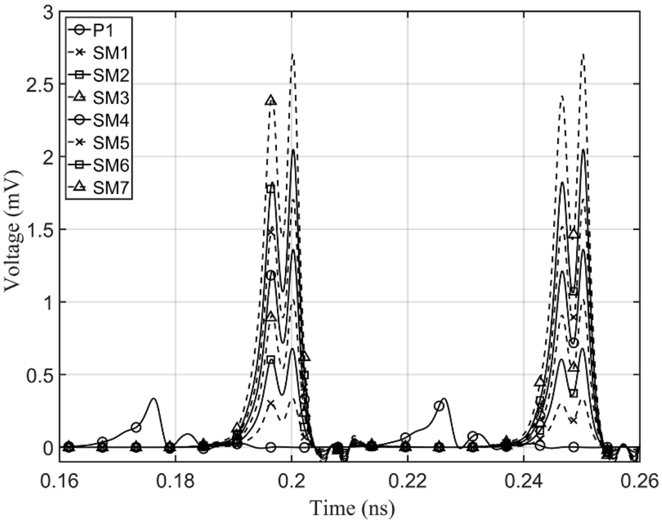

Figure 6. Simulation result of the JJ-Synapse with different weight values. The tested circuit is similar to the design shown in figure 3(e) but with 3-bit weight values. Here, we see the weight of the circuit changes as we switch the SQUID cells on and off.

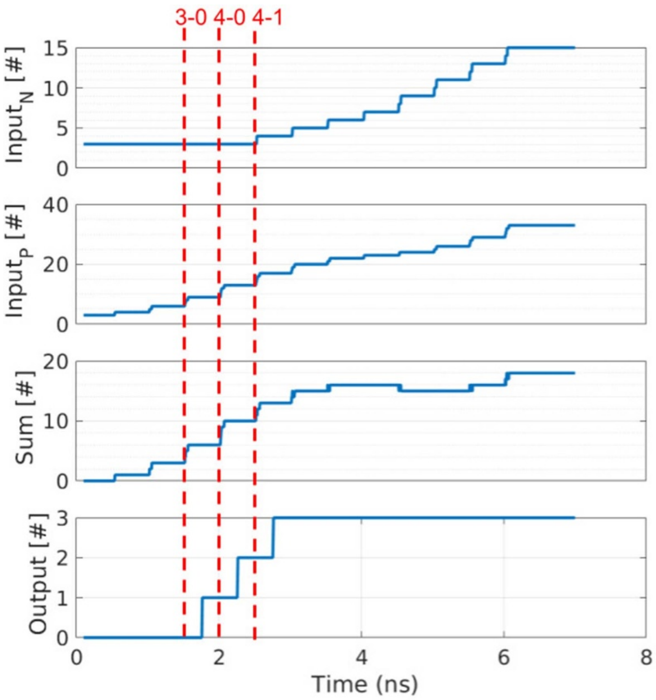

Figure 7. Simulation result of the neuron circuit with threshold 3. Here, we measure the phase and divide it by 2π to get the number of pulses. The negative and positive inputs are measured from the synapses, and the Sum is the output of the BQ circuit. The output pulse is generated when three fluxes are in the BQ circuit in less than 50 ps.

JJ-Synapses' Josephson junctions and the Sum value obtained from the BQ circuit's output. Notably, if three or more fluxons are present at the BQ circuit output within a 50ps window, the soma will generate a pulse at the JJ-Soma's output. To demonstrate the circuit's functionality, we designed a 4-input synapse with weight values of 0, 1, 2, 3 in both positive and

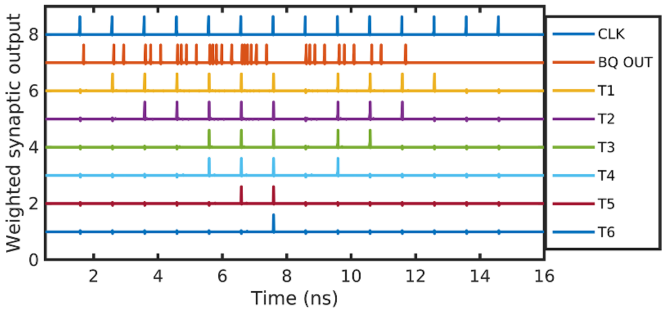

Figure 8. Simulation result for the circuit that is shown in figure 5. The T1 to T6 are the thresholds of the Soma circuit.

negative parts, effectively supporting 3-bit weights. We then interfaced this circuit to a JJ-Soma for activation function with threshold values ranging from 1 to 6. The resulting outputs are illustrated in figure 8. It's important to note that these circuits can be further stacked to accommodate 4-bit weight values and 4-bit threshold values, making them suitable for mapping different neural networks onto the presented structure. Figure 8 demonstrates the simulation results for when different synaptic weights from 1 to 6 are applied to the neurons with threshold values ranging from 1 to 6.

4. Hardware implementation

We designed the testbench layouts for a sample synapse circuit to investigate its operation. AIST CRAVITY fabricated the circuits with a 4-layer Nb process HSTP [35]. Figure 9 shows the fabricated testbench. The circuit has a DC/SFQ converter cell as input, and a splitter tree increases the number of SFQ pulses. Here, the default weight is two, which means that two of the SQUIDs always generate SFQ pulses. The weight consistently increases as the other SQUIDs are connected. Notably, in this circuit configuration on experiments, BiCMOS switches are not connected, and the SQUIDs are hardwired, meaning that they are connected to the ground by a resistor. Anticipating future advancements, a BiCMOS-based cryo-circuit will switch the SQUID connections on or off during the operation of the final circuit. The experimental setup for this circuit is similar to what we demonstrated in [36, 37].

The measurement results for a default weight of two and four are demonstrated in figure 10. Since we cannot capture an SFQ pulse, we implement the amplifier after an SFQ/DC converter circuit stage. The output that we measure on the oscilloscope is the average value of all the combined SFQ/DC pulses via the SQUID stack, and therefore, the output of 4 stages is not scaled linearly but is about 1.5× of weight 2. To confirm the functionality of CMOS switches, we designed and fabricated a SiGe-based CMOS chip and measured it in a gas transfer type GM cryocooler. The test probe, circuit layout, and measurement results of an NMOS at 4.2 K are shown in figure 11. The I_d vs. V_{ds} graphs confirm the operation of the NMOS at 4.2 K. The CMOS circuit will set the weights and will not switch during the inference of the SFQ-spiking neural network (SNN) chip. Therefore, it will not generate heat and flux noise

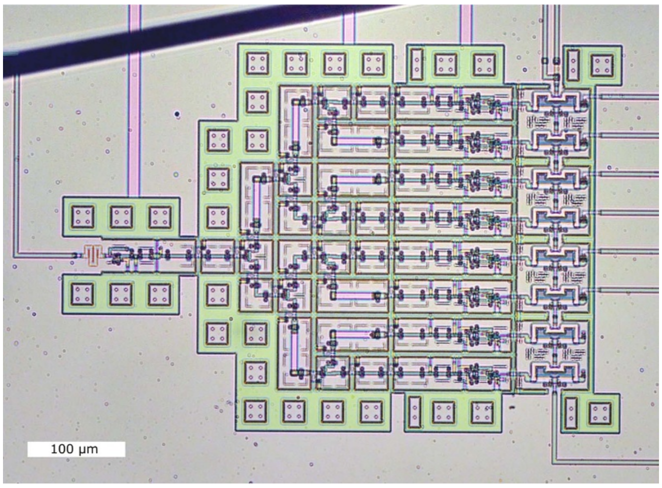

Figure 9. Test circuit layout designed with HSTP process to confirm the functionality of JJ-Synapse. The wires from SQUIDs are connected to the pads that will be wire-bonded to a Si-Ge CMOS chip for switching.

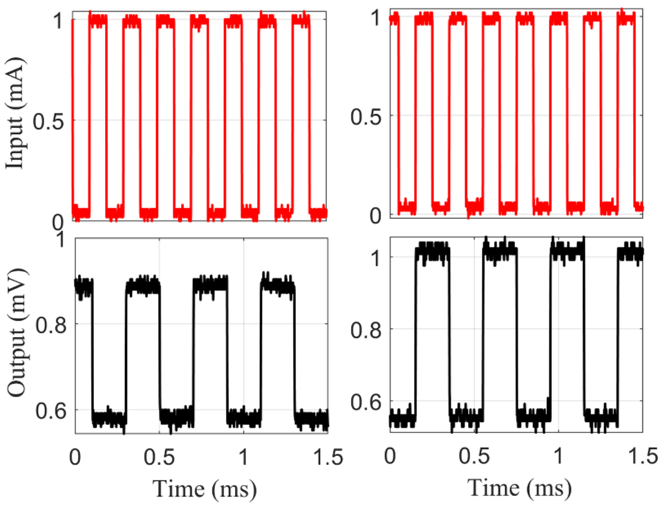

Figure 10. Measurement result for weights 2 and 4 on the left and right, respectively. Here, weight four is not scaled linearly and is 1.5× of weight 2.

for SNN operation. The applied voltage to the CMOS circuit can be lower since we are not concerned with its switching speed. Therefore, the CMOS chip can work at sub-thresholds to minimize its noise effect on SFQ circuits. The CMOS chip will sit at the top of the SFQ chip and will be bonded to it.

4.1. Hardware scalability disscusion

In the current design, each SQUID cell area is $80 \times 40\ \mu\text{m}^2$, which is twice the JTL cell. With higher weight values and a large number of synaptic connections, the current layout will not scale for larger networks. However, by improving the fabrication process, nano-SQUID implementation will be possible [38]. This can make the circuit size much smaller and scalable. The CMOS switch connection is another issue that two separate chips may cause. By fabricating the SFQ and

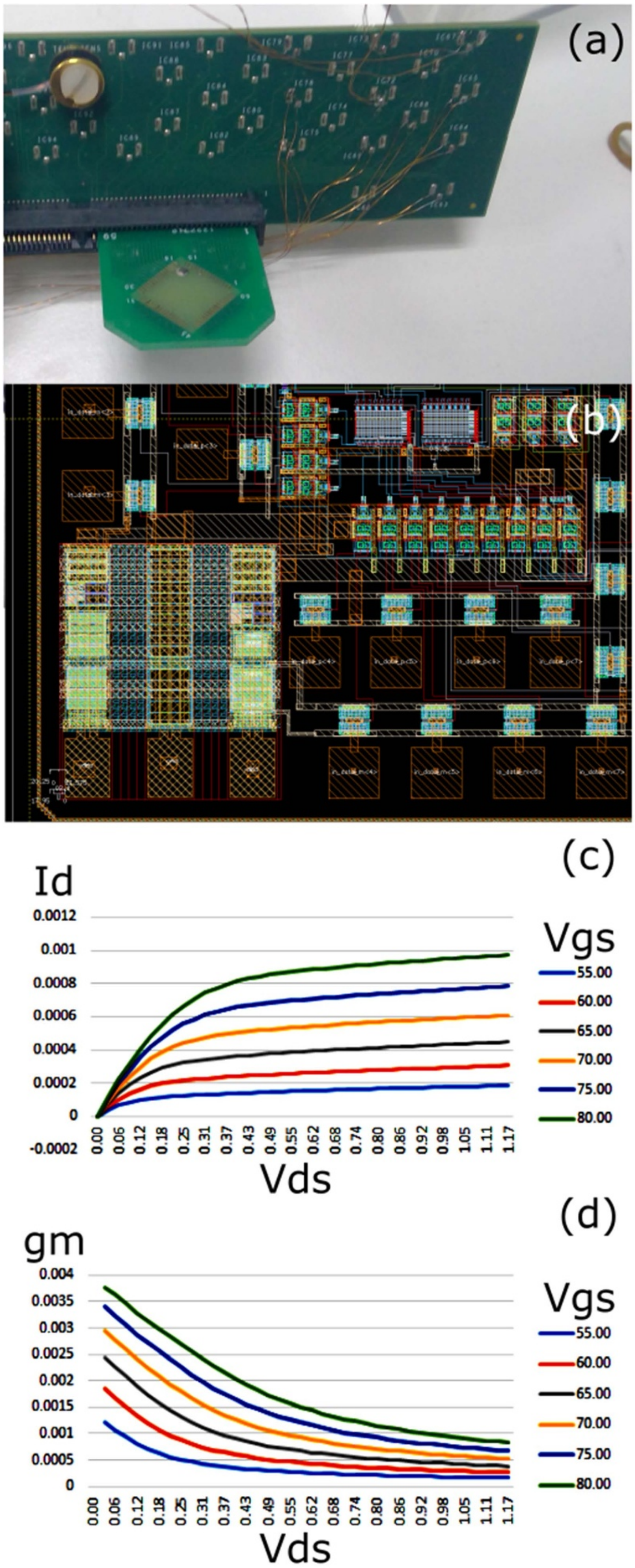

Figure 11. The SiGe cryogenic CMOS is designed for implementing the synaptic weights. (a) demonstrates the chip on the test probe that was lowered in the cryostat with the temperature sensor at the top left corner of the probe, (b) is the layout of the chip designed for IHP SiGe process, (c) is the I_d vs. V_{ds} graph at 4.2 K in different V_{gs}, and (d) is the g_m vs. V_{ds} measurements at 4.2 K in different V_{gs} values.

CMOS on the same substrate as demonstrated by Whiteley *et al* [39], it is possible to fabricate switches next to synaptic connections and directly load the synaptic weights from CMOS memory to SFQ synapses with the help of CMOS switches. This is possible by improving the fabrication process for superconductor circuits.

In existing neural network applications, the processing element (PE) consists of a MAC unit and a comparator module for the activation function. For large-scale implementations of various neuromorphic architectures, the PEs are commonly employed in a time-multiplexed manner to save area on a chip and optimize resource utilization. This time-multiplexing approach allows to efficiently share computational resources, enhancing the overall scalability of the system. Similarly, the proposed structure exhibits equivalent functionality of the MAC unit and also emphasizes reusability within the frameworks, thereby demonstrating its compatibility with such paradigms.

5. Conclusion

We have designed and fabricated a synapse circuit using a combination of superconductor and semiconductor circuit components. The synapse circuit is compatible with standard RSFQ circuits and soma cells that act as an activation function. The proposed design is suitable for realizing a SNN. The JJ-Synapse uses a series SQUID circuit with no static power consumption. The power is consumed in SFQ pulse switching and interconnects between the cells. The circuit is compact, scalable, and has an acceptable bias margin value. The resulting inference SNN will be programmable, allowing the weights to be modified after fabrication. The synapse weights are adjustable via an in-situ Si-Ge CMOS circuit. We fabricated the JJ-Synapse, and an experimental measurement was done on the pulse multiplication, confirming the circuit's functionality. The JJ-Synapse can function up to 20 GHz and have power consumption in the order of attojoule for a single operation. We simulated the whole circuit with JSIM and will test the circuit with the CMOS chip as a controller, and later, we will demonstrate the complete SNN network.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

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ORCID iDs

S Razmkhah https://orcid.org/0000-0003-0029-2800
M A Karamuftuoglu https://orcid.org/0000-0002-0951-1697
A Bozbey https://orcid.org/0000-0003-2747-310X

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